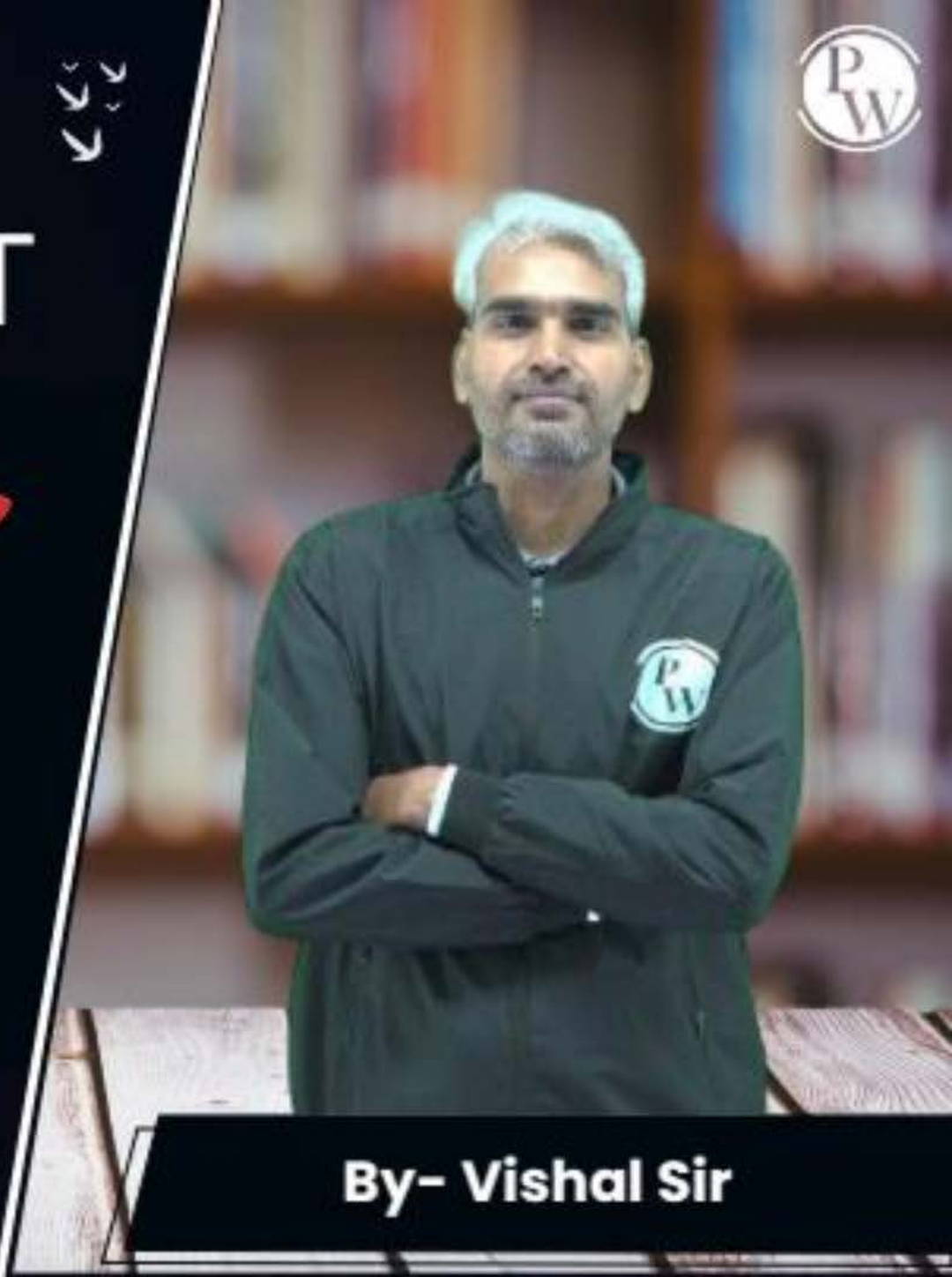
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



Topic Line covering

Topic Line independent set

Slide

Topics to be Covered



- Topic — Vertex covering
- Topic — Vertex independent set
- Topic — Connectivity

Slide

Covering

Line Covering / Edge Covering

We will try to Cover all the vertices of the graph using subset of edges

Vertex Covering

We will try to Cover all the edges of the graph using subset of vertices



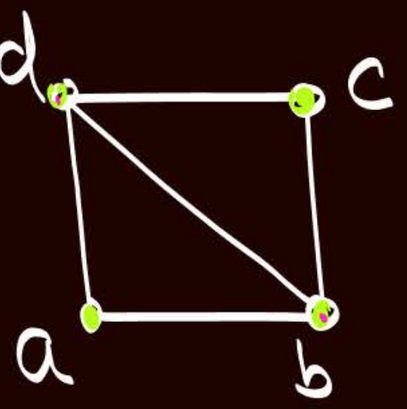
Topic : Vertex Covering

Let G=(V,E) be a graph.

A subset K of set of vertices V is called vertex covering of graph G if every edge of the graph is incident with at least one vertex of set K

{ i.e, Every edge of graph G must have at least one of its end vertex in subset K }

eg:



G

$K_1 = \{a, b, c, d\}$

$K_2 = \{a, c, b\}$

$K_3 = \{a, c, d\}$

$K_4 = \{b, d\}$

$K_5 = \{a, b, d\}$

All subsets of Vertices from K_1 to K_5 are vertex Covering of graph G



Topic : Minimal vertex covering

- A vertex covering of graph G is called minimal vertex covering if no vertex can be deleted from the set without destroying its ability to cover all the edges.

In the above example,

K_2, K_3, & K_4 are minimal vertex covering



Topic : Minimum vertex covering

/Smallest minimal Vertex Covering

- Vertex Coverings of graph G with minimum number of vertices are called minimum Vertex Coverings of graph G

In the above example,

K_4 is the minimum Vertex Covering

* Every minimum vertex covering is also a minimal vertex covering but every minimal vertex covering need not be a minimum vertex covering (eg: K_2 & K_3)



Topic : Vertex covering number (α_2)

Vertex Covering number of Graph G (α_2) = No. of Vertices in any one of the minimum Vertex Covering of G

In the above example,

$\alpha_2 = 2$

(w.r.t. K_4)

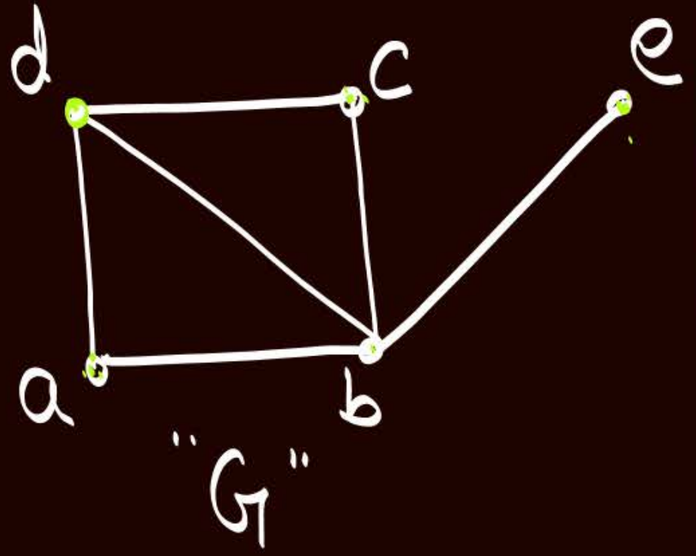


Topic : Vertex independent set

→ Let $G = (V, E)$ be a graph.

A subset 'S' of set of vertices V is called a vertex independent set if no two vertices of subset S are adjacent to each other.

eg:



$S_1 = \{e\}$

$S_2 = \{b\}$

$S_3 = \{a, e\}$

$S_4 = \{a, c, e\}$

$S_5 = \{d, e\}$

A subset of vertices with a single vertex is always a vertex independent set.

All subset of Vertices from S_1 to S_5 are vertex independent set w.r.t. graph G.



Topic : Maximal vertex independent set

- A vertex independent set of graph G is called a maximal vertex independence set if no new vertex of the graph can be added to the set without destroying its property of being independent.

In the above example.

S_2, S_4 & S_5 are maximal vertex independent set.



Topic : Maximum vertex independent set

Largest maximal vertex independent set

- Vertex independent sets with maximum number of vertices are called maximum vertex independent set of graph G

In the above example.

S_4 is maximum vertex independent set.

* Every maximum vertex independent set is also a maximal vertex independent set, but Every maximal vertex independent set need not be a maximum vertex independent set. eg. $\{S_2, S_5\}$



Topic : Vertex independence number (β_2)

Vertex independen Number (β_2) of graph G = No. of vertices in any one of the maximum vertex independent set of graph G.

In the above example,

$\beta_2 = 3$

"w.r.t. S_4"



Topic : NOTE

For any graph $G = (V, E)$

$$\alpha_2 + \beta_2 = |V(G)|$$

(Vertex Covering no. of graph G + Vertex independence no. of graph G) = Total no. of Vertices in graph G

Find α_1, β_1, α_2 & β_2 for the following graphs

(i) Complete graph K_n

(ii) Cycle graph C_n

(iii) Wheel graph W_n

(iv) Complete bipartite graph $K_{m,n}$

(V) Star graph with 'n' vertices

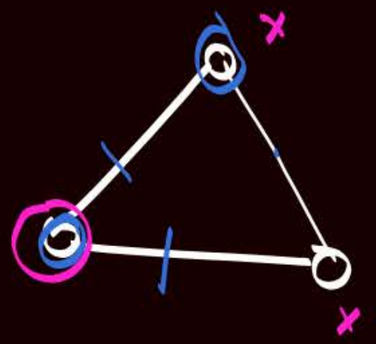
α_1 = line covering no.

β_1 = line independence no.

α_2 = Vertex covering no.

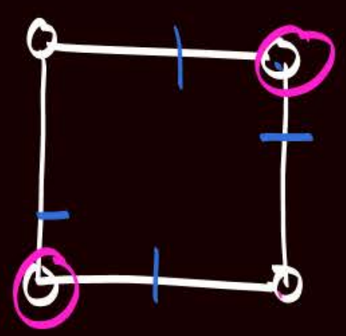
β_2 = Vertex independence no.

	Line Covering No. α_1	Line independence No. β_1	Vertex Covering No. α_2	Vertex independence No. β_2
* ① Complet graph (K_n)	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$	$(n-1)$	1
② Cycle graph (C_n)	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$
③ Wheel graph (W_n)	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$	$1+\lceil \frac{n-1}{2} \rceil = \lceil \frac{n+1}{2} \rceil$	$\lfloor \frac{n-1}{2} \rfloor$
④ Complete Bipartite graph ($K_{m,n}$)	$\text{Max}(m,n)$	$\text{Min}(m,n)$	$\text{Min}(m,n)$	$\text{Max}(m,n)$
⑤ Star graph with n-vertices	$(n-1)$	1	1	$(n-1)$



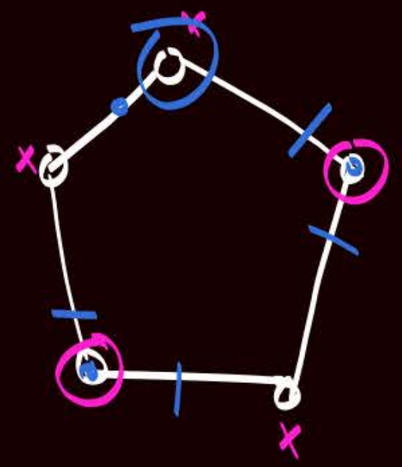
$\alpha_2 = 2$

$\beta_2 = 1$



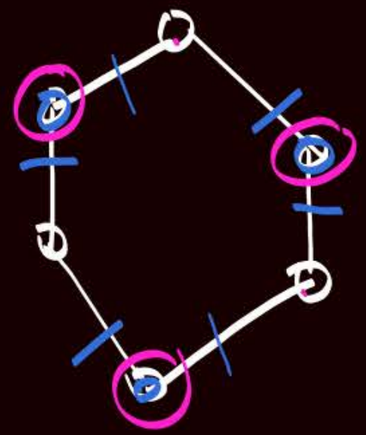
$\alpha_2 = 2$

$\beta_2 = 2$



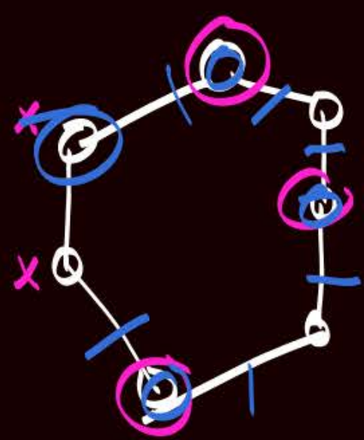
$\alpha_2 = 3$

$\beta_2 = 2$



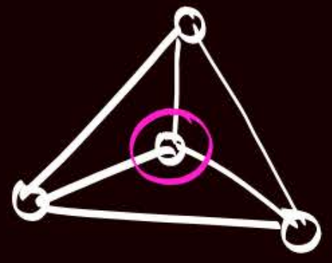
$\alpha_2 = 3$

$\beta_2 = 3$

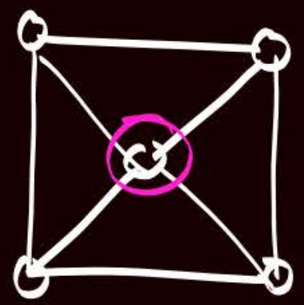


$\alpha_2 = 4$

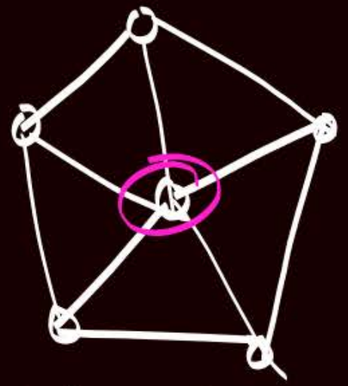
$\beta_2 = 3$



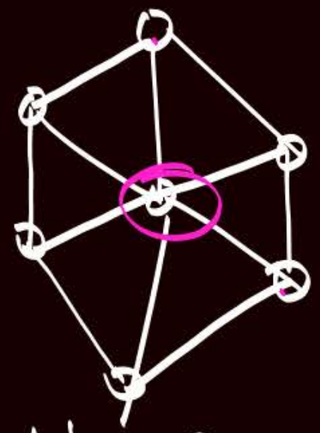
$W_4 = C_3 + hub$



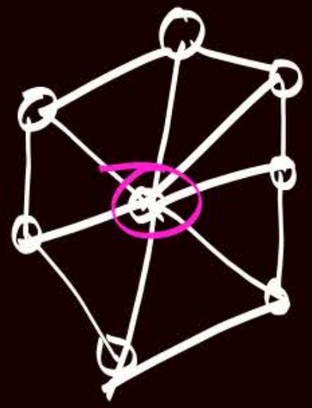
$W_5 = C_4 + hub$



$W_6 = C_5 + hub$



$W_7 = C_6 + hub$



$W_8 = C_7 + hub$

$$\beta_2(W_n) = \beta_2(C_{n-1}) = \left\lfloor \frac{n-1}{2} \right\rfloor$$

$$\alpha_2(W_n) = hub + \alpha_2(C_{n-1}) = 1 + \left\lceil \frac{n-1}{2} \right\rceil = \left\lceil \frac{n+1}{2} \right\rceil$$

it will cover all spokes of wheel



Topic : NOTE

- For a graph $G = (V, E)$

Similar statements can not be defined w.r.t. Line Covering & Line independent set.

(1) If 'S' is a vertex independent set of graph G, then 'V-S' is a vertex covering of graph G.

{ if S is maximal then 'V-S' is minimal
if S is maximum then 'V-S' is minimum }

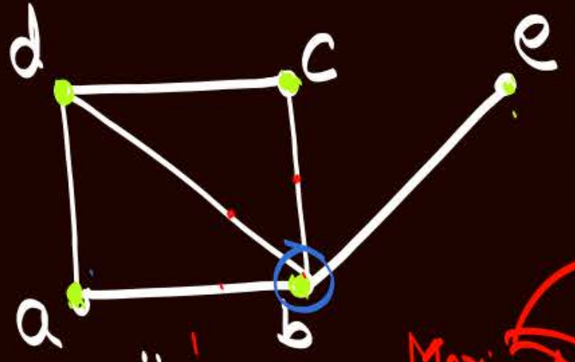
and

(2) If 'k' is a vertex Covering of graph 'G' then 'V-k' is a vertex independent set of graph G.

{ if 'k' is minimal, then 'V-k' is maximal
if 'k' is minimum, then 'V-k' is maximum

Relationship b/w Vertex Covering & Vertex independent set

eg:



"G"

$G = (V, E)$

$V = \{a, b, c, d, e\}$

Vertex independent Set (S) = Vertex Covering $(V-S)$

Vertex independent Set (S)	Vertex Covering $(V-S)$
$S_1 = \{e\}$	$V-S_1 = \{a, b, c, d\}$
$S_2 = \{b\}$	$V-S_2 = \{a, c, d, e\}$
$S_3 = \{a, e\}$	$V-S_3 = \{b, c, d\}$
$S_4 = \{a, c, e\}$	$V-S_4 = \{b, d\}$
$S_5 = \{d, e\}$	$V-S_5 = \{a, b, c\}$

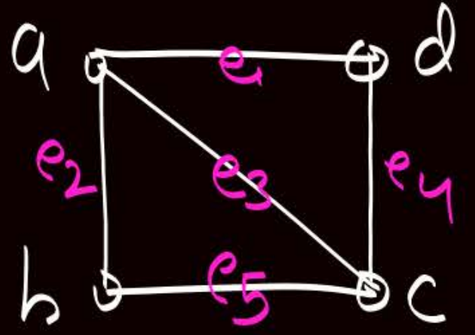
Maximal

"Maximum"

Minimal Vertex Cover

Minimum

Let $G = (V, E)$ be a graph



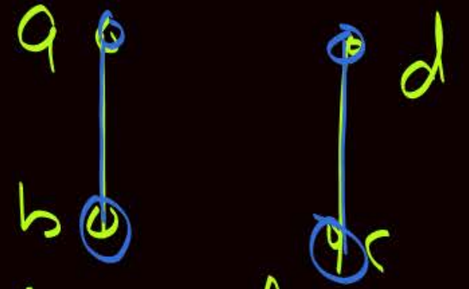
$G = (V, E)$

$V = \{a, b, c, d\}$

$E = \{e_1, e_2, e_3, e_4, e_5\}$

We can observe

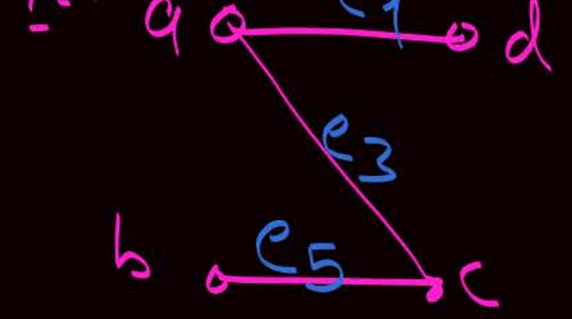
$C_1 = \{e_2, e_4\}$



is a line Covering

But $E - C_1 = \{e_1, e_3, e_5\}$

i.e.



is not an independent set of edges.

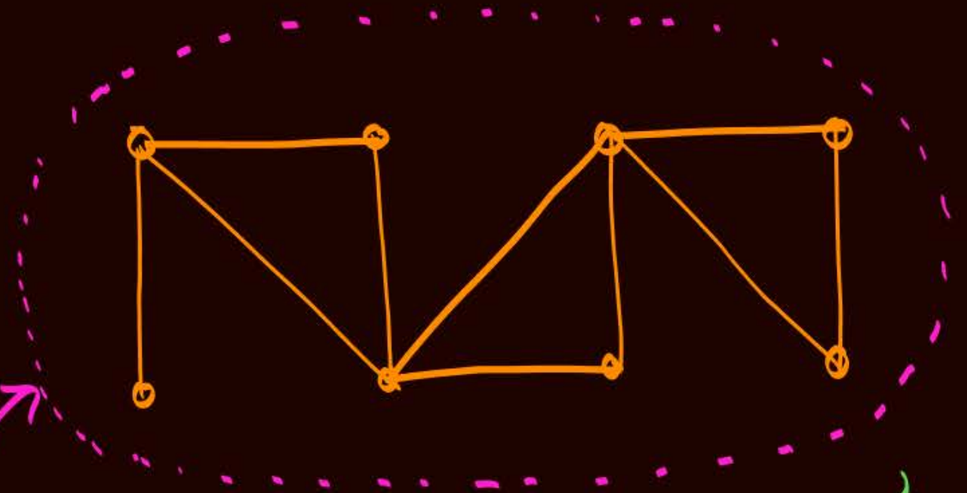
Connectivity



Topic : Connectivity

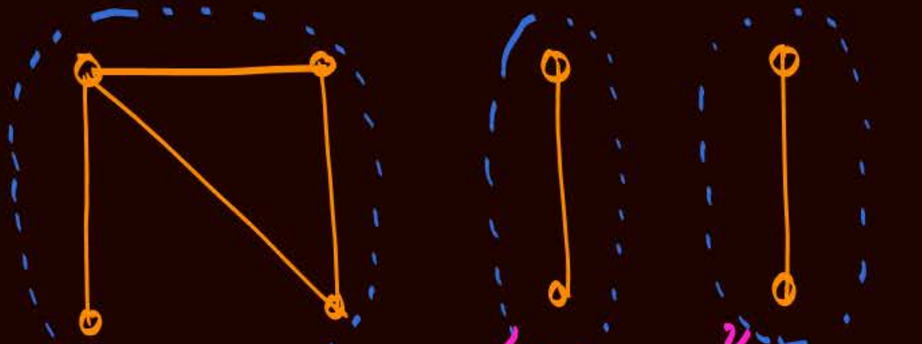
A graph is said to be connected if there exists a path between every pair of vertices.

If a graph (with 2 or more vertices) is not connected, then it will have two or more connected components.



Single Connected Component

Connected graph



Three Connected Components

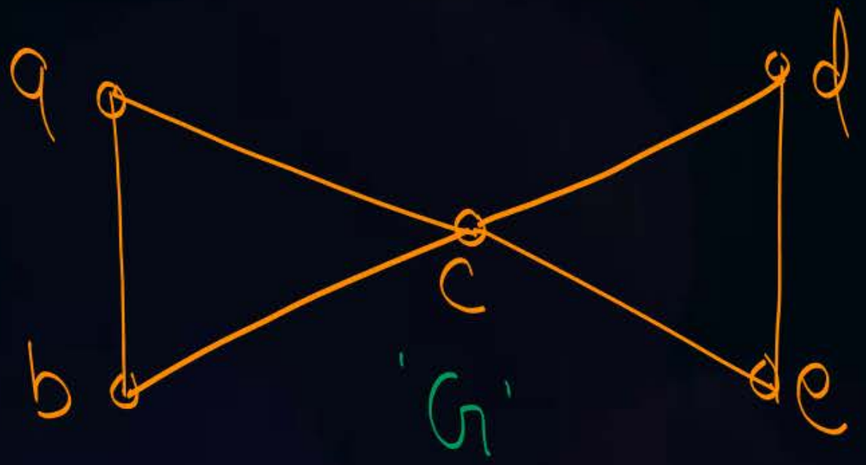
disconnected graph



Topic : Cut-vertex

"Articulation point"

Let G be a connected graph, a vertex $v \in G$ is called a cut vertex if deletion of vertex "v" from graph G makes the graph disconnected.



delete 'c'



it is a Connected graph

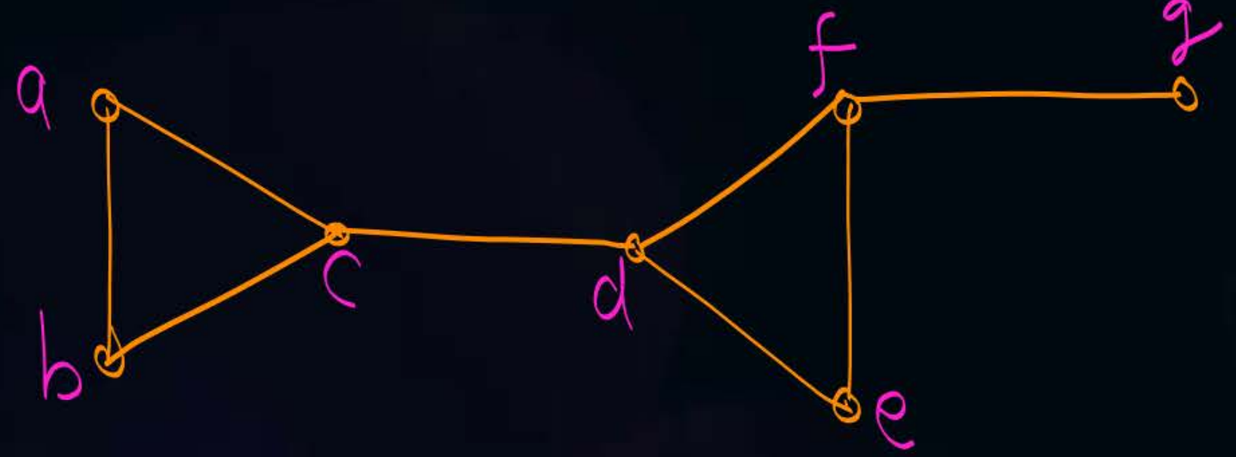
Graph becomes disconnected

∴ 'c' is a Cut-vertex



Topic : Cut-edge / Bridge

Let G be a connected graph, an edge $e \in G$ is called a cut edge or bridge if deletion of edge "e" from graph G makes the graph disconnected.



Connected graph

Edge $\{c,d\}$ is a Cut edge as well as Edge $\{f,g\}$ is also a Cut edge



Topic : NOTE

① An edge of a connected graph G can be a cut-edge (bridge) if and only if that edge is not a part of any cycle in the graph.

② At least one vertex associated with a cut-edge is a cut-vertex of that graph

eg: w.r.t. cut-edge {c,d} in above example, both c & d are cut-vertex

w.r.t. cut-edge {f,g} in above example, 'g' is not a cut-vertex but 'f' is a cut-vertex

Note:- Existance of Cut-vertex in a graph does not imply the existance of Cut-edge, but

Existance of a Cut-edge in a Graph implies the existance of Cut-vertex.



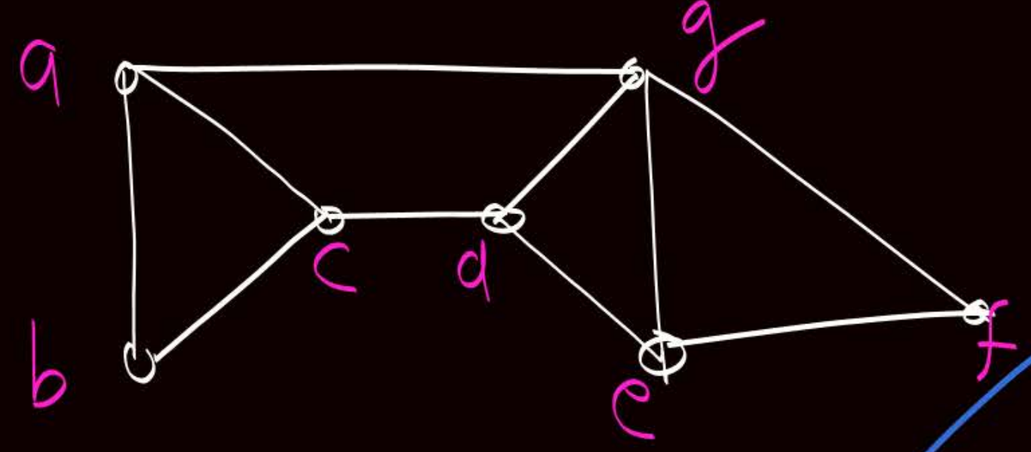
Topic : Cut-set

"No edge can be removed"

A minimal set of edges in graph G, such that removal of all the edges of that set from the graph G makes the graph disconnected is called cut-set of graph G.

& connected

If we leave any edge of that set and delete the remaining edges then it should never disconnect the Connected.



The set of edges $\{\{c,d\},\{d,e\},\{a,g\}\}$ is a Cut set or not?

Removal of all the edges can make the graph disconnected, but it is not a Cutset, because it is not minimal.
(We can remove the edge $\{d,e\}$ from the set)

Here $\{\{c,d\},\{a,g\}\}$ is a Cut-set.



Topic : Vertex connectivity

- Let G be a connected graph, the minimum number of vertices whose removal from graph G makes the graph disconnected is called vertex connectivity of graph G.
Vertex connectivity of a graph G is denoted by $\kappa(G)$

- Vertex connectivity of graph is 1 if and only if cut-vertex exists in the graph.

- If graph G is disconnected graph, then $K(G) = 0$



Topic : Edge connectivity

Let G be a connected graph, the minimum number of edges whose removal from graph G makes the graph disconnected is called edge connectivity of graph G.

Edge connectivity of a graph G is denoted by $\lambda(G)$

- Edge connectivity of graph is 1 if and only if cut-edge exists in the graph.
- If Graph G is a disconnected graph, then $\lambda(G) = 0$

Note:-

① For a connected graph G,

$$\lambda(G) \leq \delta(G) \quad \text{---eq}^{n}\ (1)$$

② For a connected graph G

$$K(G) \leq \lambda(G) \quad \text{---eq}^{n}\ (2)$$

By eq^{n} ① & ②

$$K(G) \leq \lambda(G) \leq \delta(G)$$

H.W: Find Vertex Connectivity ($K(G)$) and Edge Connectivity ($\lambda(G)$) For the following graphs.

	Vertex Connectivity	Edge Connectivity
① Complete graph K_n		
② Cycle graph C_n		
③ Wheel graph W_n		
④ Complete bipartite graph $K_{m,n}$		
⑤ Star graph with n-vertices		

H.W: {already done}

#Q. If G is a forest with n vertices and k connected components, how many edges does G have?

H.W. [MCQ]

#Q. In a connected graph G if we delete an edge then no. of components are

A 0 or 1

B 1 or 2

C 2

D ≥2

[MCQ]

HW

#Q. In a connected graph G with 'n' vertices if we delete a vertex then number of components are

A ≤ 2

B 1 or 2

C Lies between 1 and n -1

D ≥2

H.W.

#Q. A simple graph with n vertices is necessarily connected if number of edges are more than E, then find the value of E.

H.W.

#Q. Maximum number of edges possible in a simple undirected graph with n-vertices and 'k' components ?



2 mins Summary

Topic Vertex covering

Topic Vertex independent set

Topic Connectivity

Slide

THANK - YOU